Solutions and teaching notes for chapter 2

Teacher use. The answers below concern the tasks not worked in the lesson. First have students state their method; give graded help before the solution. The durations are estimates to be calibrated in class.

Diagnostic and remediation

1. $x^2 - 6x + 9$. *Action:* Rebuild the product of two binomials.
2. $(x-4)(x+4)$. *Action:* Use the difference of two squares on two examples.
3. $x = 2$ or $x = -5$. *Action:* Have students say aloud the zero-product rule and its hypothesis.
4. $x = -3$ or $x = 3$. *Action:* Distinguish the positive square root from the solutions of the equation.
5. Negative before 3, zero at 3, positive after 3. *Action:* Draw the line, then make the sign table.
6. 4, reached only at 1. *Action:* Start from the fact that a square is non-negative.

Investigation activity

For width x, the length is $12 - x$; the geometric domain is $(0, 12)$. The requested areas are respectively 20, 32, 36, 32 and 20 square units. For every admissible width, $A(x) = x(12-x) = 36 - (x-6)^2 \leq 36$, with equality only at $x = 6$. Thus the maximum-area rectangle is a square of side 6. Trials suggest the result; the nonnegative square supplies the proof.

Immediate applications

1. $g(x) = (x+1)^2 - 4 = (x-1)(x+3)$. Roots 1 and -3, minimum -4 at -1.
2. The first is $3(x-1)^2 = 0$, so 1. For the second, $\Delta = -16 < 0$: no real solution.
3. $f(x) = a(x-1)(x-4)$; $4a = -8$, so $f(x) = -2(x-1)(x-4)$.
4. $(x-5)(x+1) < 0$ gives $(-1, 5)$. The vertex of $x(8-x)$ is at 4, outside the domain; on $[0, 3]$ the function is increasing and its maximum is 15 at 3.

Exercises: method and results

Exercise 2.1: Completing the square. $x^2 - 8x + 7 = (x-4)^2 - 9 = (x-1)(x-7)$. The leading coefficient is positive; the quadratic is negative or zero between the roots: $[1, 7]$.

Exercise 2.2: Discussing without heavy calculation. $x^2 + 4 > 0$ always, no root. $(x+2)^2 \geq 0$, double root -2. For the last one, roots -4 and 1; positive on $(-4, 1)$, negative outside, zero at the endpoints.

Exercise 2.3: Choosing a strategy. Special product: $x = \pm 3$. Sum and product: 4 and 5. For the third, no obvious integer root: $\Delta = 17$ and $x = \dfrac{3 \pm \sqrt{17}}{4}$. The discriminant works in every case, but is not always the most efficient.

Exercise 2.4: Recovering a parabola. $f(x) = a(x+2)(x-3)$. At 1: $-6a = 12$, so $a = -2$; $f(x) = -2x^2 + 2x + 12$, hence $f(0) = 12$.

Exercise 2.5: A vegetable garden against a wall. The two widths use $2x$ metres; length $20 - 2x$. Domain $[0, 10]$ for the function, $(0, 10)$ for a non-degenerate rectangle. $A(x) = 20x - 2x^2 = 50 - 2(x-5)^2$. Maximum 50 m^2, dimensions 5 m and 10 m.

Exercise 2.6: A parameter. $\Delta = 4 - 4m$. Two roots if $m < 1$; double root 1 if $m = 1$; none if $m > 1$. The parameter m stays fixed when we solve for x.

Exercise 2.7: Link with sequences. $u_n = (n-1)(n-4)$; negative for $1 < n < 4$. Since n is an integer, the only indices are 2 and 3. The set of indices is not the whole real interval.

Exercise 2.8: Delayed review. The minimum is -5 at -3. The inequality gives $(x+3)^2 \leq 1$, that is $-1 \leq x+3 \leq 1$, so $[-4, -2]$.

CQFD, exercise 43 p. 99, questions a and c. a: obvious root 1, product -30, so the second root is -30. c: write $4x^2 + 9x + 5$; root -1, product $\frac{5}{4}$, so the second root is $-\frac{5}{4}$. Check the sums, -29 and $-\frac{9}{4}$ respectively.

CQFD, exercise 49 p. 99, questions a and b. The numbers are the roots of $t^2 - st + p$. a: $t^2 - t + 1$, discriminant -3: no real pair. b: $t^2 + 60t + 756$, discriminant $576 = 24^2$: -18 and -42. Their sum is -60 and their product 756.

MCQ: answers and reasons

MCQ 1.

1. C. Set each factor to zero.
2. A. $4 - 12 = -8$.
3. D. The square vanishes at 2, value 5.
4. B. $-\frac{b}{a} = \frac{6}{2} = 3$.
5. C. Positive leading coefficient, strict inequality.

MCQ 2.

1. D. Special product.
2. A. $\frac{c}{a} = -\frac{2}{3}$.
3. C. The function increases up to 2; on this domain, right endpoint 1.
4. B. For a nonzero, a zero discriminant characterises the double root.
5. D. The multiplying coefficient is free and nonzero.

True or false

1. False: $x^2 + 1$ is strictly positive.
2. False without a hypothesis on a; $-(x-1)(x-2)$ is positive between 1 and 2.
3. True: the square of a real number is positive or zero; equality only at 2.

Differentiated teaching. For a technical block, offer only the first step (domain, factor or reference sample space), then withdraw the help. For an independent student, ask for a second method and the treatment of a limiting case. Orally, allow an initial reformulation in the language the student has mastered, then require a precise mathematical sentence in French.